Probability and Statistics (MT-2005)

Semester Project - Spring 2026

FAST NUCES Islamabad

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Deadline: 10th May 2026, 11:53 PM
The project can be done by, at most, a pair of students.

Submission Guidelines:
Submit your work in a `.zip` file containing the `.ipynb` file and the report in PDF format.
Naming convention: `rollnum1_rollnum2_section.zip`
Example: `i230030_i230750_G.zip`

Linear Regression

Background and Motivation

In the digital media landscape, understanding what drives user engagement is critically important for publishers, marketers, and content creators. Predicting the popularity of an article before or shortly after publication can help organizations optimize content strategy, improve audience targeting, and increase reach. The shared dataset provides an opportunity to explore how different characteristics of an article - such as its length, sentiment, topic, and publishing time - affect its likelihood of being widely shared.

More on the Dataset

The UCI Online News Popularity Dataset contains information about online news articles collected from a content aggregation platform. Each instance represents a published article and includes a wide range of features describing its content, structure, and publication context. These features include metrics such as the number of words in the article, number of images and videos, keyword statistics, sentiment polarity, subjectivity scores, and timing of publication.

The target variable is the number of times an article was shared on social media platforms (referred to as “shares”), which serves as a proxy for the article’s popularity. The dataset contains dozens of input variables (mostly numerical), making it suitable for multivariate analysis and regression modeling.

Project Requirements

You are required to:

- Preprocess the data and apply EDA, as demonstrated in previous coding projects
- Normalize or standardize the data if needed
- Identify any redundant features
- Attempt to explain any unusual correlations
- Encode variables where needed
- Understand relationships between features using metrics like correlation
- Describe the data using statistical measures studied in class
- Build a linear regression model and interpret its coefficients
- Check assumptions of linear regression (multicollinearity, outliers, etc.)
- Evaluate model performance
- Classify the target variable (shares) into High and Low classes
- Evaluate classification using accuracy and confusion matrix
- Analyze regression coefficients and their magnitude
- Interpret the effect of each feature on the target variable
- Identify the most significant predictors of article popularity
- Discuss potential multicollinearity issues (if present)
- Visualize distributions of key variables (including one heatmap)
- Plot relationships between selected features and the target variable

- Perform hypothesis testing on 5 selected features, including:
  - Formulating null and alternative hypotheses
  - Applying appropriate statistical tests
  - Interpreting results in context

Model Evaluation: Mean Baseline vs. Fitted Models

In predictive modeling, there is an ongoing debate in statistics regarding whether it is more appropriate to fit a model (such as linear regression or classification algorithms) or to rely on simple baseline approaches like using the average value of the target variable. In this project, using the mean number of shares serves as a baseline model against which more complex models can be compared. You are required to evaluate whether a fitted model significantly improves prediction performance over this simpler approach. Based on your findings, argue which approach is more suitable for this dataset, supporting your reasoning with appropriate performance metrics and analysis.

All information on the dataset has been provided in the `.names` file.

Deliverables

- Complete Python code in `.ipynb` format
- A report describing methods, results, and conclusions
- Implementation screenshots and generated plots

Note: During the demo, you will be evaluated on your understanding of the dataset, the problem, and your implementation.

Dataset Source

K. Fernandes, P. Vinagre and P. Cortez. *A Proactive Intelligent Decision Support System for Predicting the Popularity of Online News.* Proceedings of the 17th EPIA 2015 - Portuguese Conference on Artificial Intelligence, September, Coimbra, Portugal.